

Analysis of SDR type and settings for best OGN ground station reception distance

June 2026 by Wojtek SP9WPN

TL; DR

RTL-SDR Blog version 3 device showed better reception performance and should be preferred over version 4 for setting up OGN receiver stations unless there are compelling reasons to use v4. These reasons include strong **unfiltered** out-of-band signals, but it's way more efficient to install a filter.

There's no meaningful difference between “silver” and “black” v3s, despite different tuner chips.

Simple noise floor scan can show defects in the SDR's radio path.

For best reception distance, even with LNA in front, use highest possible input gain value (accepting occasional input overdrive). `OGN.MinNoise=7.0` and `OGN.MaxNoise=12.0` seem to be good choices for ogn-rf config. Don't just go for nice looking spectrogram. Occasional input overdrives do not matter in the big picture.

Avionix Engineering's openAir multitrack reception distance is as good as RTL-SDR v3, but device seems to suffer from dropping random packets.

Script I have used are published on: https://github.com/sp9wpm/ogn_rx_tuning_scripts/

Test setup

At the beginning, my goal was just to compare RTL-SDR Blog's models v3 and v4. I created a test setup where two receivers worked in parallel, fed with identical RF signals. Since OGN radio traffic is not repeatable, this allowed a qualitative comparison of two devices. As results came in and surprised me, I introduced also quantitative approach methods.

Both devices were installed in parallel with same remaining hardware and software. During the test period, receivers were swapped places few times thus all individual cables, USB and splitter ports worked both signal paths. There were no observable differences after swaps.

Other checks were performed to confirm the results were not caused by external reasons:

- replacing each SDR with 50Ω standard dummy load to check if their presence does not cause interference in the other SDR,
- terminating both SDR inputs with 50Ω dummy loads to check internal noise for artifacts and roughly measure noise floor,
- powering Bias-Tee from external 5V DC source instead of Raspberry's USB port,
- completely removing Bias-T and LNA/SAW from the path.

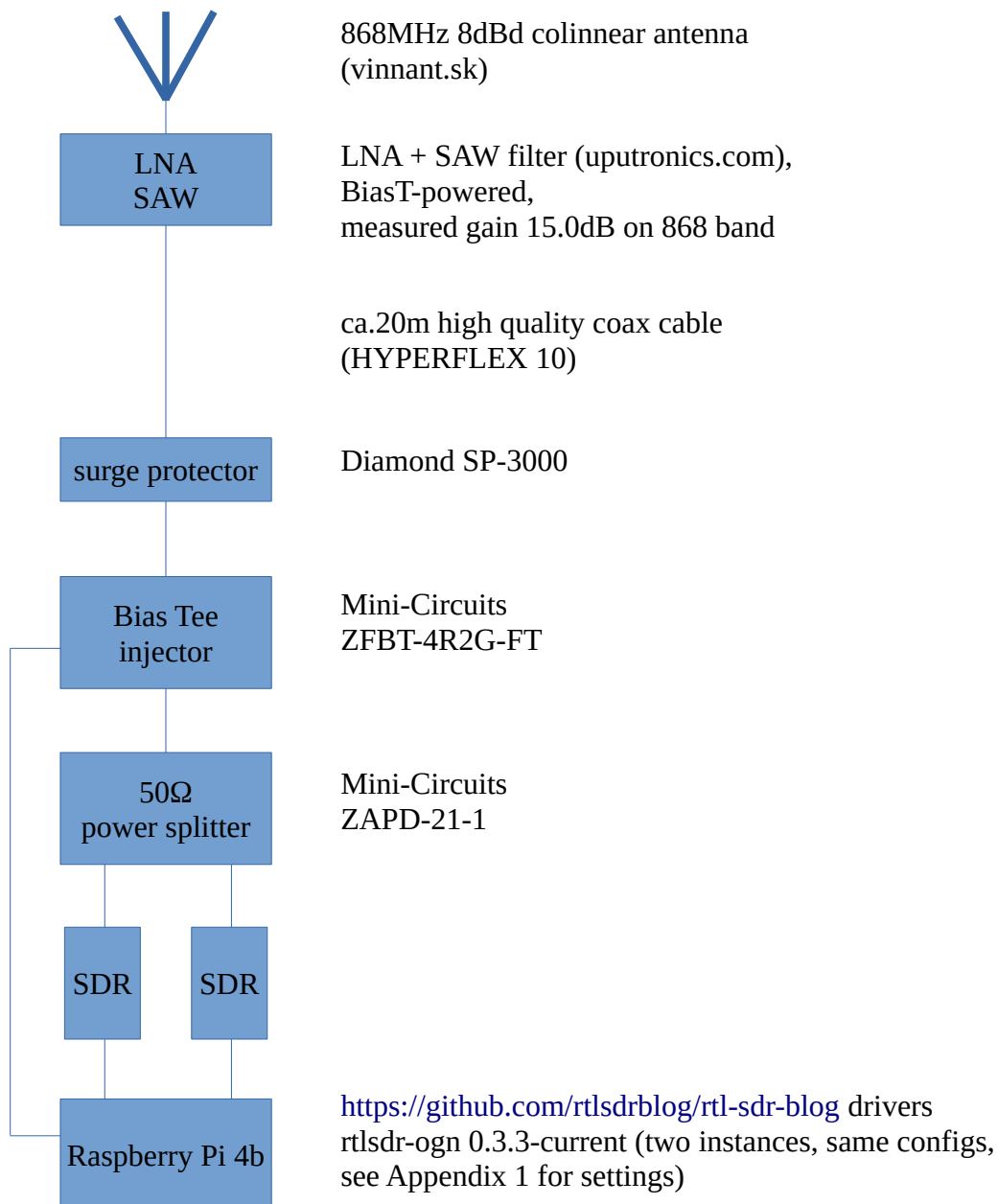
Clock calibration (using *gsm_scan*) was done on each tested device, all errors were below 5 ppm.

In total, three v3 devices were tested (two “silver”, one “black”) and two v4 devices. Avionix Engineering multitrack receiver was added to the test pool at the end.

All devices were purchased at different times in the official web shops, so risk of counterfeit hardware is minimal.

External (noisy!) environment

Tests were performed at Gliwice Airfield (EPGL) in March-May 2026. The airfield is located at the edge of big large urban area with lots of non-OGN signals on 868 band. Many of them seem to ignore european duty cycle requirements. Strongest VHF FM station is 4,5km away with 2KW ERP. The worst RF-troublemaker is a cellphone base station using 810 MHz just 300m away.



Test target

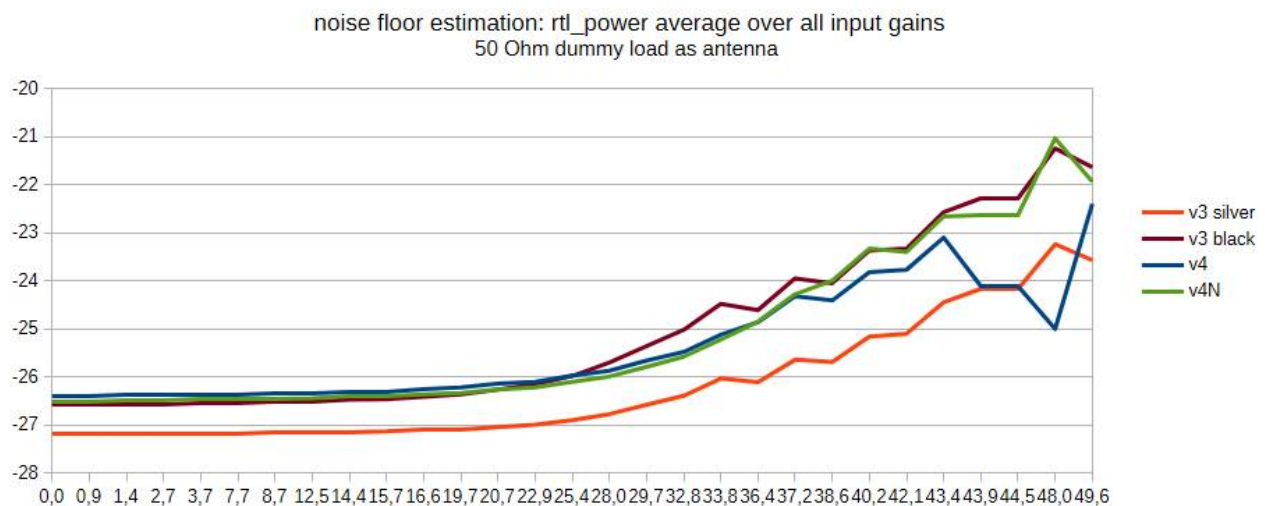
A test target was installed – an OGN tracker placed few kilometers from the airfield with -10 dB attenuator before the antenna. This was considered a “static weak target”, convenient for quantitative approach.

Methodology

During the testing period, I changed and introduced methods. I describe them all here, not in actual chronological order.

Noise floor test

“Poor-man's noise floor measurement” (code on github) was made using *rtl_power* was run at each possible local gain with 50Ω-terminated antenna port. One v4 device (the first tested) shows unexplained signal loss (observed later also with live signals) on three high gain levels:



Another v4 unit (v4N) was obtained and didn't show this phenomenon. It's likely that the specific device was broken. The device was disqualified from further testing, although it seemed to work fine with other gain settings.

This test can therefore be recommended as fast screening of a SDR receiver.

Distance and SNR comparison (live traffic)

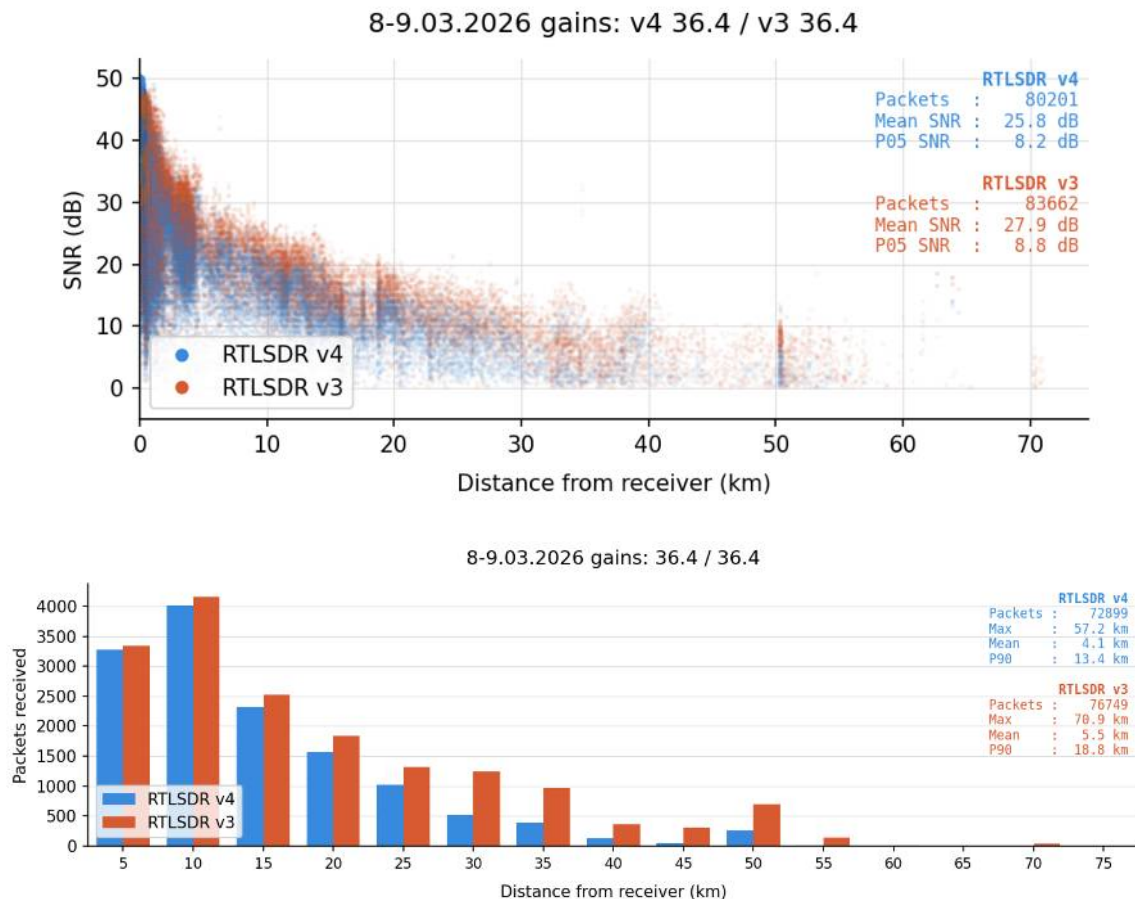
Output of *ogn-decode*'s monitoring port (default 50001) was logged on both instances. This data contains all decoded OGN and Flarm frames with target positions (especially distance to receiver) and measured SNR. Other signals (LoRa mesh networks, weather stations) are not reported¹.

Relayed frames were rejected as well as frames with “parity correction” flag (question mark at the end of entry).

¹ This has a small drawback: only GFSK modulated signals were used in the test.

First test with “auto-gain” offered by OGN-decode showed unexpected and significant differences between SDRs. I decided to perform separate comparison sessions (lasting usually a single day each) with different **local input gains** of both devices to check influence of this setting. Gains were then set manually and auto-gain feature of *ogn-rf* was disabled².

Two chart types were produced. First one shows distribution of SNR (calculated by receiver software) with distance. Second shows total number of frames decoded from distance groups. While SNR is more of technical value, I consider maximum reception distance as ultimate indicator of station performance.

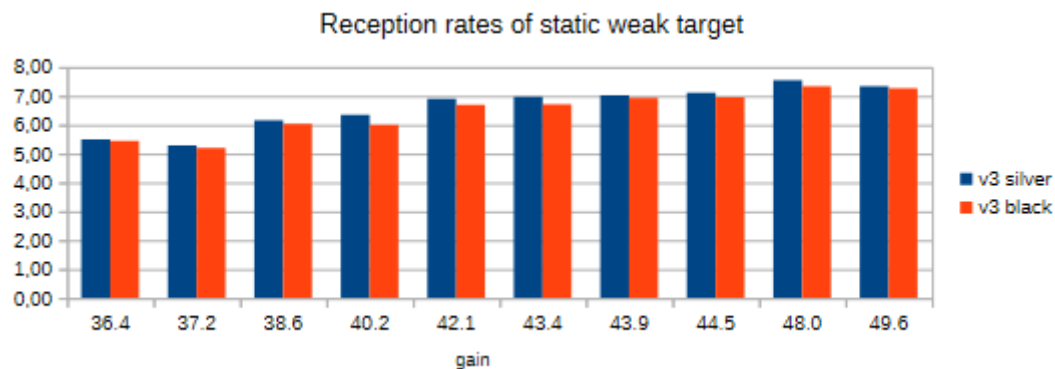


Both charts are consistent, so I focused on distance groups, which is easier to read.

² by setting OGN.MinNoise=-20.0 and OGN.MaxNoise=50.0

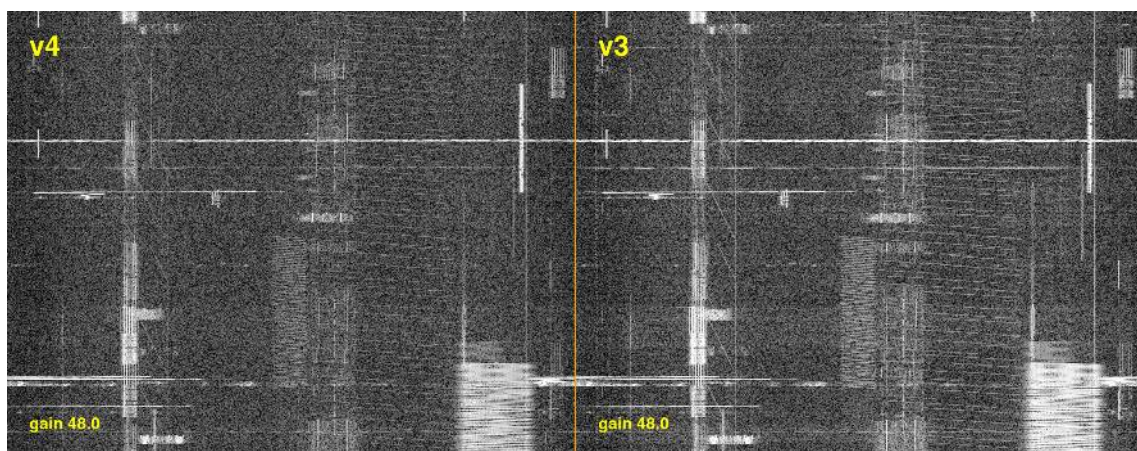
Static weak target packet count

Using a test target allowed for quantitative analysis without waiting weeks for averages to settle. Receivers were cycled through a range of input gain values (each step lasting a minute or two) and the number of received packets from the target was counted. Day long session was enough for reasonable and stable data across many gain settings.



Spectrograms

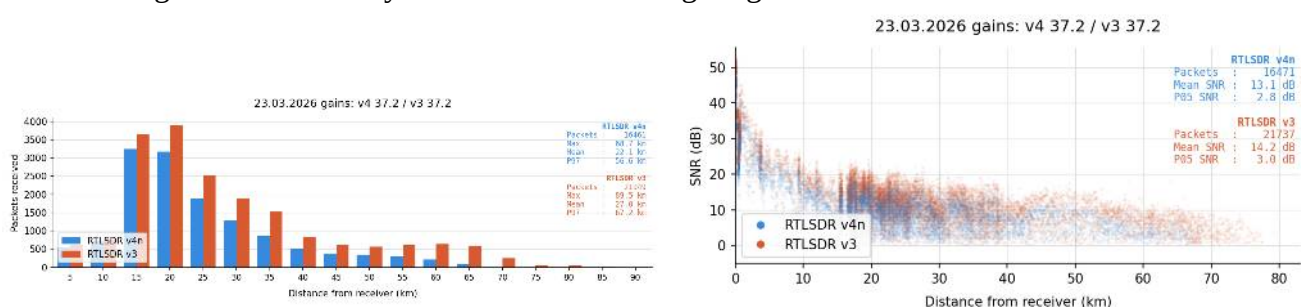
Spectrograms from OGN-rf were collected alongside the other tests. They help to confirm that test is conducted properly and assess or compare level of noise, overdrives etc. as seen by the decoder. A script allowed fetching two spectrograms at same time, so they are very easy to compare visually.



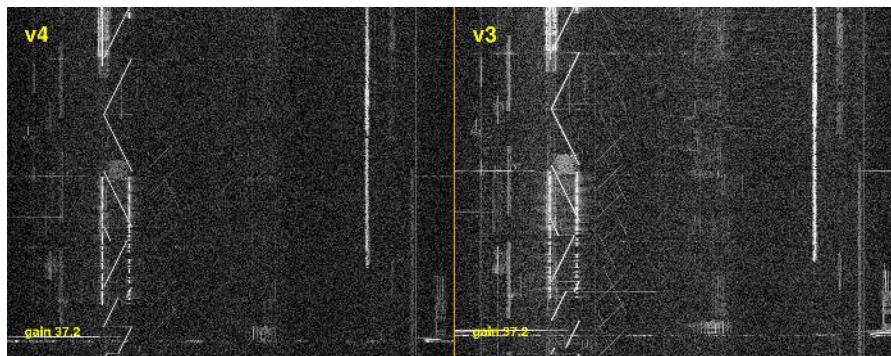
Results

RTL-SDR v3 vs RTL-SDR v4

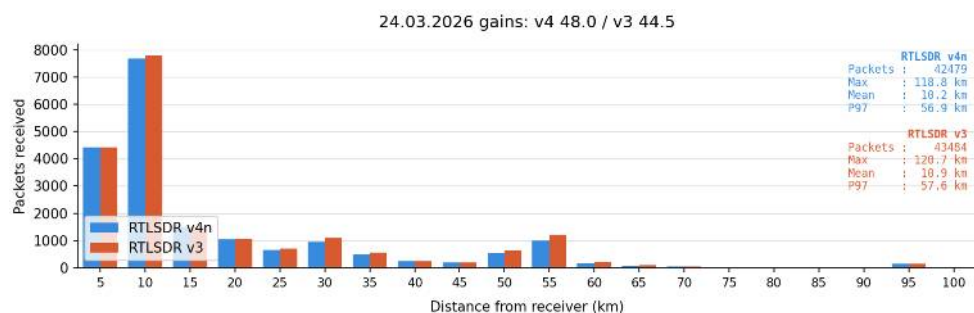
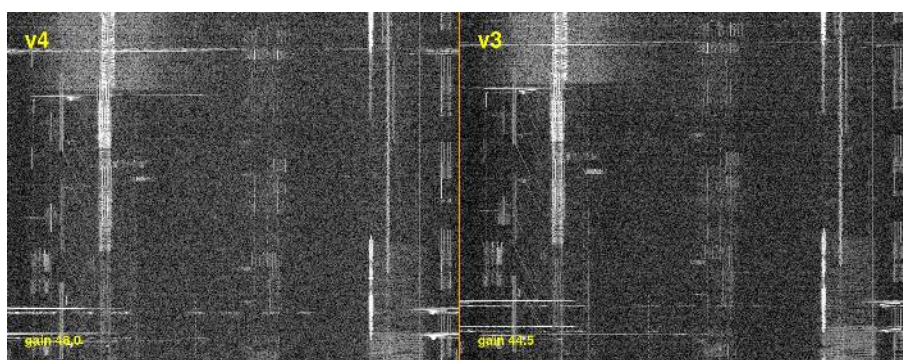
The first surprise came very early: while input gain settings for these two devices are same (in dB), observed signal levels are way different and decoding range confirms that:



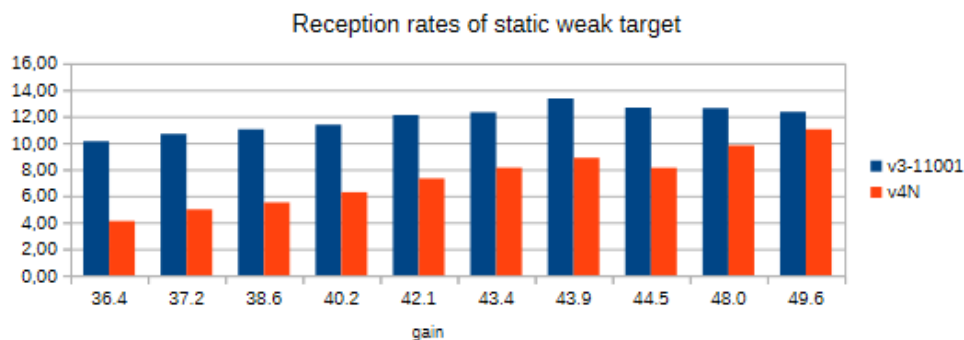
Spectrograms shows that v4 apparently lacks contrast and weak details at moderate gain level:



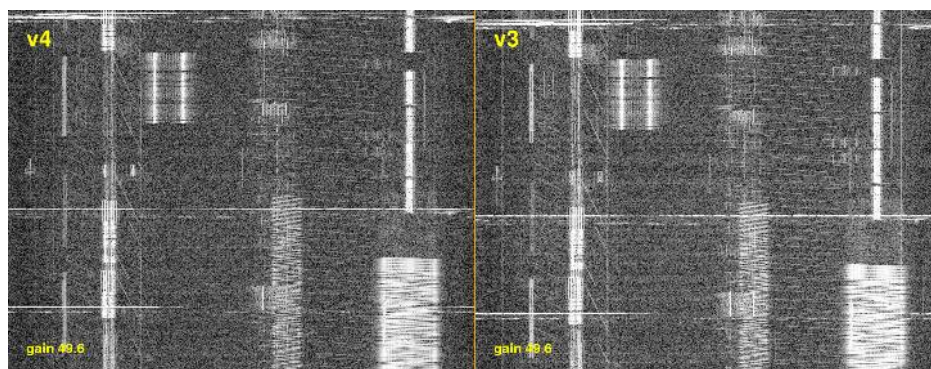
After many iterations, I found pair of local gains where spectrograms came visually closest to each other at 48.0 (v4) and 44.5 (v3) gains, but received packets counts still slightly favored v3:



“Static weak target” test proves that v4 “likes” high gains, despite spectrograms looking ugly. RTL-SDR v3 performs much better, showing a plateau where gain changes result in minimal differences:

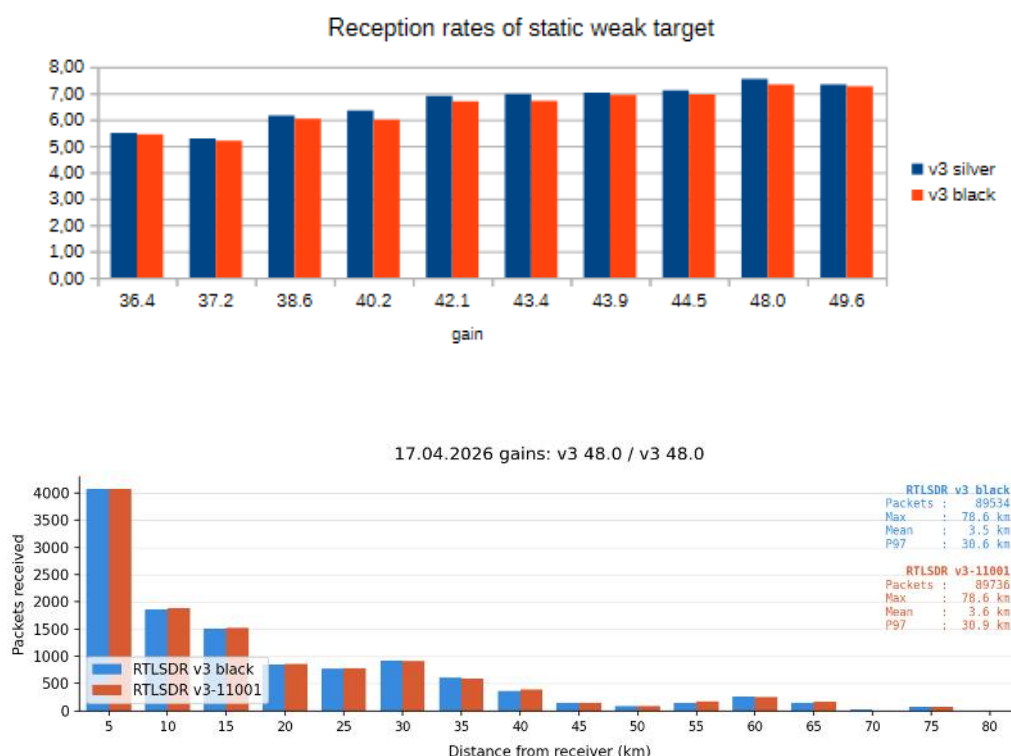


It should be noted that such high gain settings seem unreasonable as incoming signals have upfront 10 dB amplification³. Occasional amplifier overdrives were observed here.



RTL-SDR v3 “Silver” vs RTL-SDR v3 “Black”

RTL-SDR does not offer “Silver” v3 receivers anymore, all new products are in a black case. Color is not the only change: radio tuner chip changed from R820T to R860. I obtained one of them to check if this influences the performance. Apparently it does not:

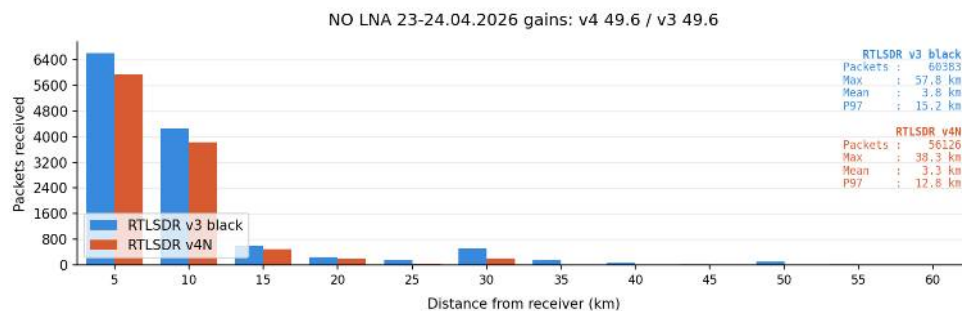


Unfiltered, non-amplified input

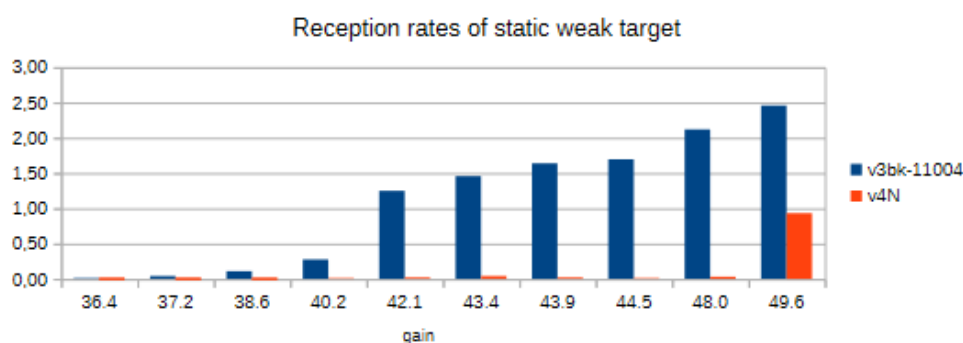
There is a common belief that having a LNA “under the antenna” amplifies received signals to a level where typical imperfections of the latter signal path are negligible. This theory did not hold water in my tests. I decided to completely remove LNA (integrated with SAW filter) and let the receiver cope with natural RF as it comes from the antenna. Since there's cellular base station just

³ +15 dB LNA+SAW, -3 dB splitter, -2 dB (estimated) cables and connectors

300m away using 810MHz, all receivers responded very badly to it. While v3 performed better, both were just poor and intriguingly required **maximum** possible gain (49.6 dB).

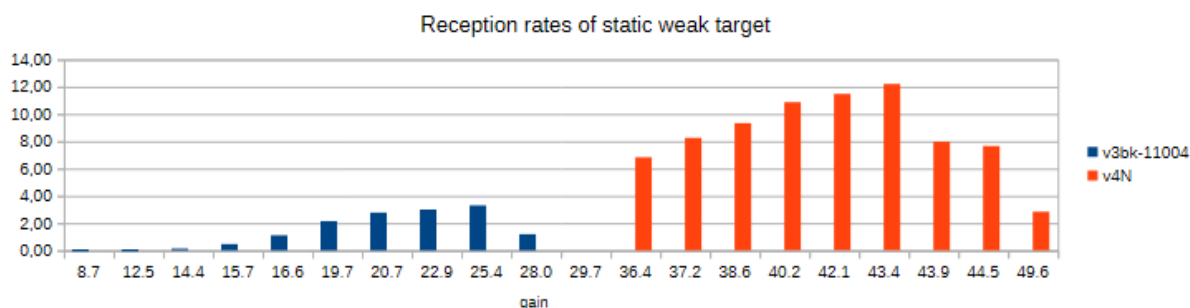


My test target was too weak for v4 device in this scenario:



Non-filtered input

I installed another, general-purpose LNA (namely LNA4ALL, measured gain 18.3 dB). The strong signal easily overdrove v3's tuner, allowing only low values of gain to be used, still with awful results:

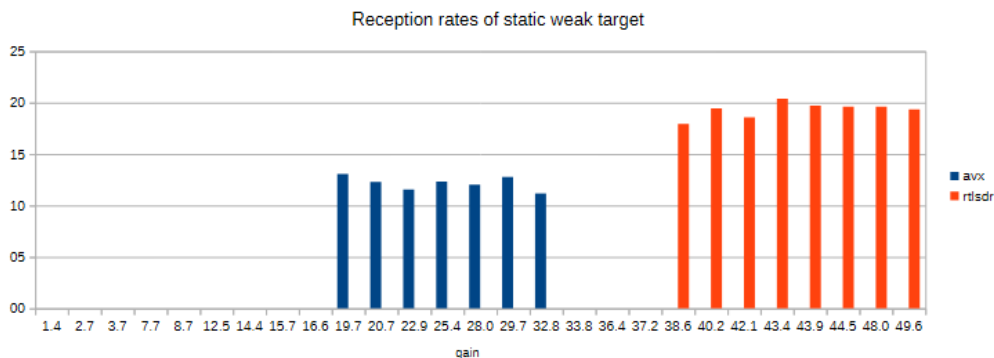


This was the only time where v4 unit outperformed v3. My hypothesis is: v4 has more band filters. While they are useless in this setup (main unwanted signal at 810 MHz), their insert losses might have saved the day. Version 3 device clearly capitulated.

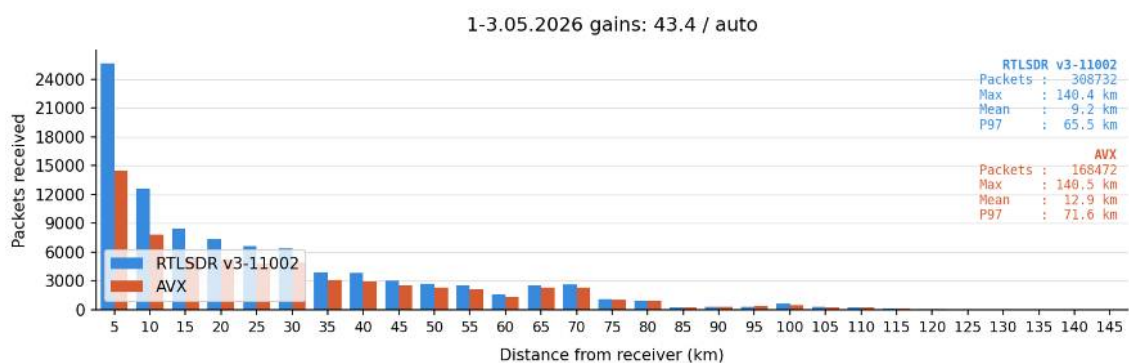
Avionix multitrack vs RTL-SDR v3

Lastly, after installing back the original LNA and SAW filter, I tested openAir multitrack receiver from Avionix Engineering (I will now use AVX to mention it). Deep inside is the same RTL2832U chip, but the signal path is different. OGN decoding software seems to be the same as publicly available rtl-sdr-ogn binaries, but there may be some proprietary additions. I was curious to check how good it is and whether it is worth the price as a “gold standard” decoder.

Apparently there's an extra amplifier at the input start, as useful gain values are much lower. At first, AVX seemed clearly inferior to RTL-SDR v3:



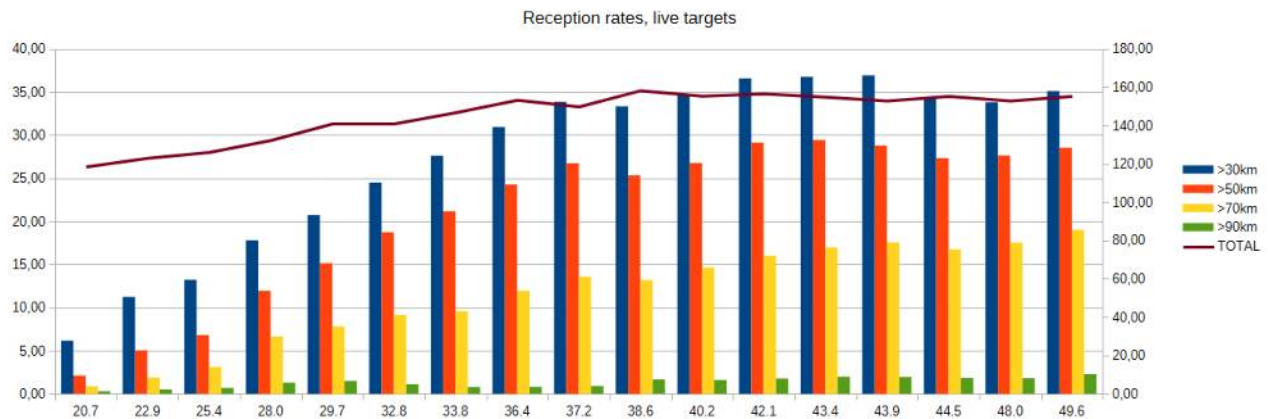
Yet looking at reception ranges shows more context:



AVX was able to decode packets from same (really long) distances as RTL-SDR, but with lower packet numbers. This may have a few explanations, but most plausible for me is dropped RF data due CPU overload. AVX uses Pi Zero and is also a powerful ADS-B 1090 decoder, thus its system load rarely drops below 5 (where “4” means all 4 CPU cores are saturated).

Long-distance reception (live data)

Finally, I restored installation to the production setup (single SDR – v3 “Black”, Bias Tee and splitter removed, test target disabled) and tested it for over a week for reception of long distance (DX) packets. This approach counted received packets from over 30, 50, 70 and 90 km. Local gain settings were cycled as previously. As this method relies heavily on local traffic, long averaging was necessary. Fortunately World Gliding Championships nearby provided a solid test pool.



Results confirmed that using a “static weak target” was a valid approach. High gain settings are welcome and deteriorate stronger (closer) signals only slightly. This test does not require any hardware setup – only a good amount of patience. I would recommend it for fine-tuning OGN receiver sites. Scripts are published on https://github.com/sp9wpm/ogn_rx_tuning_scripts/

Main conclusions

RTL-SDR v4 underperforms

RTL-SDR v4 devices are marketed with improved filtering, better out-of-band signal separation and lowered thermal footprint. While MDS (minimum discernible signal) is a bit worse in manufacturer's test⁴, this should have no effect on pre-amplified signal.

Version 3 shows a desired, comfortable “plateau” of optimal gain settings.

Few quotes found on the Internet describe v4 as having higher noise floor and worse reception results. My tests seem to confirm these opinions.

Auto-gain feature of ogn-rf, recommended new settings

Default rtl-sdr-ogn installation has auto-gain feature enabled. It adjusts SDR's input gain to keep measured noise within limits, which are set 2 to 6 dB. The idea is sound on paper, but my tests showed that allowing higher noise levels (around 10 dB) offer the best reception results, at least in my noisy RF environment.

⁴ <https://www.rtl-sdr.com/rtl-sdr-blog-v4-dongle-initial-release/>

Final thoughts

I tried to rule out any possible external influences. Despite that, v3 devices performed much better without an obvious reason. Manually setting SDR input gains allowed to fine tune reception performance in both cases, but did not make receivers even.

Further testing in lab environment could be done, but I leave it to the device manufacturer. Both devices are available on the market with similar prices.

Apart from concluding that v4 devices are simply not recommended for 868MHz weak signals, there are still possible, while unlikely, explanations:

- bug in rtl-sdr drivers,
- inconsistency between performance of individual units (only one v3 and two v4 devices were tested),
- local RFI or resonance on v4's internal clocks/oscillators (coming from Raspberry Pi itself or a nearby device),
- unknown/undetected methodology error,
- a RF-quiet environment can bring different results and conclusions.

Appendix 1 – used rtl-sdr-ogn settings

```
RF.BiasTee = {set per device, between -4 and -1}  
RF.FreqCorr = 0;  
RF.SampleRate = 2.0  
RF.FreqPlan = 1  
# RF.GSM calibration disabled (commented out)  
RF.OGN.Gain = {tested variable}  
Demodulator.DetectSNR=11.0 dB  
Demodulator.ScanMargin=50.0 kHz
```